Sergio Márquez

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Senior Software / Product Engineer | Ruby on Rails, JavaScript, React & PostgreSQL

Summary

Senior Software / Product Engineer with 15+ years building production web applications, specializing in Ruby on Rails and PostgreSQL. Led architecture and technical planning for a six-engineer team and the incremental modernization of a 500,000+ line Rails codebase. Improved page-load times by 20% to 80% in data-heavy workflows. Product-minded engineer who takes features from user research through production, including a shipped teacher-facing LLM feature using structured output.

Skills

- **Languages & Frameworks:** Ruby, Ruby on Rails, JavaScript, React, Hotwire, Hotwire Native, Tailwind CSS
- **Data & Backend:** PostgreSQL, Redis, Sidekiq, API development, AWS, Stripe
- **Testing & Delivery:** RSpec, Capybara, test-driven development, CI/CD, CircleCI, Datadog, performance optimization
- **Architecture & Product:** System design, technical planning, product discovery, user research, LLM integration, structured output

Work Experience

Senior Product Engineer (Ruby on Rails), Contract May 2024 — Present
KaiPod Learning (Y Combinator S21) *Boston, MA / Remote*

- Build and ship full-stack features for Newton, a Ruby on Rails platform available to every school in KaiPod's Partner Network of 150+ schools across 30+ states, within a four-developer product engineering team.
- Work directly with microschool founders and teachers to identify operational problems, design solutions, and carry them through to production.
- Shipped a production teacher-facing LLM task generator using structured output to create student tasks from teacher-selected topics; teachers can select, regenerate, or edit suggestions before use.
- Expanded real-time messaging with Turbo Streams over Action Cable, including search, reactions, read receipts, threaded conversations, and pagination; improved messaging performance through query optimization and layered caching.
- Implemented Stripe API and webhook integrations for recurring monthly billing and customizable per-student plans.

Lead Software Engineer Aug 2021 — May 2024
Ruby on Rails Engineer Jul 2017 — Aug 2021
Decisiv, Inc. *Glen Allen, VA / Remote*

- Led architecture and technical planning within a six-engineer team, mentored junior engineers, and reviewed and approved most pull requests.
- Used Datadog APM to identify bottlenecks in complex workflows, reducing affected page-load times by 20% to 80% through bounded queries, N+1 elimination, lighter JavaScript, and asynchronously loaded React widgets.
- Cut CircleCI workflow time from approximately 40 minutes to 8 minutes through test-suite modernization, Capybara updates, flaky-test reduction, redundant-spec removal, and parallelized execution.
- Led the incremental modernization of a 500,000+ line Rails codebase from Rails 4.0 and Ruby 1.8 to Rails 7.0 and Ruby 3.1, using dual booting and staged testing without major downtime.
- Migrated complex AngularJS and jQuery widgets to asynchronously loaded React and vanilla JavaScript while retaining a server-rendered architecture; redesigned and rebuilt product applications and third-party integrations.

Ruby on Rails Engineer, Full-time May 2016 — May 2017
Paper Kite Ltd (PaperKite) *Wellington, New Zealand / On-site*

- Developed resilient Ruby on Rails APIs for mobile applications, including BPme, and built data-analysis tools for clients while integrating external payment services.

Earlier experience: ThinkCERCA (Ruby on Rails Engineer, Feb 2015 - Apr 2016), DevSpark (Mid Senior Ruby on Rails Engineer, Jun 2013 - Feb 2015), SoftPro (Junior Full-Stack Software Developer, Sep 2012 - Jun 2013), and Freelance Full-Stack Web Developer (Aug 2011 - Sep 2012).

Education

Universidad Tecnológica Nacional Mar del Plata, Argentina
Bachelor of Science in Computer Science *Jul 2009 — Dec 2014*

Languages

Spanish: Native | English: Full professional proficiency